SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

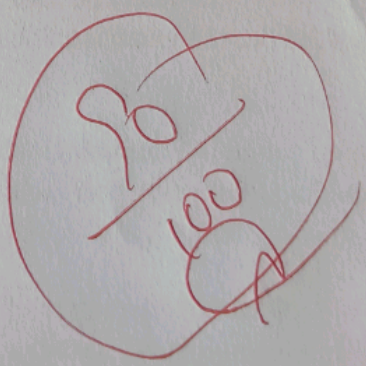
APPLICATION- BASED QUESTIONS

Code of Conduct:

I G. Ram Srinivas *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

G. Ram

Signature of the student:



Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

   a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
   b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

   a. Deduce the most likely original message.
   b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

   a. Demonstrate how the encryption process transforms the message step-by-step.
   b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

   a. Analyze how investigators could detect the presence of hidden data within the image.
   b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

   a. Determine the most probable original message by reversing the encryption process.
   b. Critically evaluate why such a method is vulnerable in modern communication systems.
   c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25 20	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30 25	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20 20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15 15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10 10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

90

CO1 - AT-1

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course:- CSA5101- cryptography and Network security.

1A) Event communication security - caesar cipher:-

plain text:- MEET AT NOON

A fixed shift of 4 positions is applied. this is a caesar cipher

a) Enayption:-

plaintext:- M E E T A T N O O N

shift +4 :- Q I I X E X R S S R

ciphertext :- Q I I X E X R S S R.

plain text:- MEET AT NOON

ciphertext :- QI IX EX RSSR

b. Decryption:-

An authorized member knows that the encryption uses a shift of 4. therefore, they shift every ciphertext letter 4 positions backward.

For example:

Q I X E R S

M E T A N O

The caesar cipher is simple and easy to implement, but it is not secure for modern communication because there are only 25 possible non-zero shifts.

2. Intercepted suspicious Message - Atbash cipher:-

given ciphertext:- GSRH RH Z HVXIVG

The structure strongly suggests the Atbash cipher, where each alphabet letter is replaced by its reverse counterpart.

The mapping starts as:-

plain text	A	B	C	D	E	F	G	H	I	J	K	L	M
cipher text	Z	Y	X	W	V	U	T	S	R	Q	P	O	N
plain text	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
cipher text	M	L	K	J	I	H	G	F	E	D	C	B	A

a. Decryption:-

GSRH

G → T

S → H

R → I

H → S

GSRH → THIS

RH

R → I

H → S

RH → IS

Z → A

HVXIVG

H → S

V → E

X → C

I → R

V → E

G → T

GSRH RH Z HVXIVG

THIS IS A SECRET

b. Justification:-

The message has several charactristics of an Atbash cipher

1. RH is a two letter word. After Reverse-alphabet substitution it becomes IS, which is a common English word

Z becomes A, another very common word.

The six-letter word HVXING decrypts to SECRET, which forms a meaningful sentence.

4. Atbash uses a fixed substitution relationship

5. The plaintext structure becomes:

THIS / IS / A / SECRET

conclusion the most likely original message " THIS IS A SECRET".

3 secure messing system:-

A company uses a keyword-based encryption technique to encrypt the word HELLO using the keyword key.

a) Demonstrate the encryption process:-

the vigenere cipher encrypts each plaintext character using a corresponding character from a keyword unlike the caesar cipher, the shift changes according to the keyword letters

plaintext = Hello

keyword = key

the keyword contains only three letters and the plaintext contains five letters, repeat the keyword until both strings have the same length.

plaintext	H	E	L	L	O
Repeated keyword	K	E	Y	K	E

the replaced keyword is KEYKE

Apply the encryption formal

$$c = (p+k) \bmod 26$$

position 1 $(7+10) \bmod 26 = 17 \rightarrow R$

position 2 $(4+4) \bmod 26 = 8 \rightarrow I$

position 3 $(11+24) \bmod 26 = 9 \rightarrow J$

position 4 $(11+10) \bmod 26 = 21 \rightarrow V$

position 5 $(14+14) \bmod 26 = 18 \rightarrow S$

ciphertext = RIJVS

b. Explain how the repeating keyword influences each character of the ciphertext :-

The keyword determines the shift applied to each plaintext letter. Because the keywords is repeated, each character is encrypted accordings to the corresponding key coracter.

For Example:-

H+K = R : K produces a shift of 10 positions.

E+E = 1 : E produces a shift of 4 positions.

L+K = V : The repeated K applies a shift of 10 positions again.

The decryption formula is

$$P = (C - K) \bmod 26$$

The receiver repeats the same keyword KEYKE and subtract each key value from the corresponding ciphertext value This recovers the original message HELLO

·security analysis The Vigenere cipher is stronger than a simple caesar cipher because it uses multiple shifts Now a short keyword creates patterns that can be exploited through cryptanalysis. modern secure messing should use established encryption algorithms.